(i) Gradient mask.

$z_1 \quad z_2 \quad z_3$

$z_4 \quad z_5 \quad z_6$

$z_7 \quad z_8 \quad z_9$

$G_x = z_8 - z_5$ and $G_y = z_6 - z_5$

$\nabla f = [G_x^2 + G_y^2]^{1/2} \Rightarrow$

$[(z_8 - z_5)^2 + (z_6 - z_5)^2]^{1/2}$

$\nabla f \approx |z_8 - z_5| + |z_6 - z_5|$

(ii) Robert cross gradient operator

$G_x = z_9 - z_5$ and $G_y = (z_8 - z_6)$

$\nabla f = [G_x^2 + G_y^2]^{1/2} = [(z_9 - z_5)^2 + (z_8 - z_6)^2]$

$\nabla f \approx |z_9 - z_5| + |z_8 - z_6|$

(iii) Sobel operator

$G_x = (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)$

$G_y = (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)$

$\nabla f = |G_x| + |G_y|$

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \quad \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

single variable fourier

$$F(u) = \int_{-\infty}^{\infty} f(x)\, e^{-j2\pi ux}\, dx$$

$$f(x) = \int_{-\infty}^{\infty} F(u)\, e^{j2\pi xu}\, dx$$

Descrete function

$$F(u) = \frac{1}{M} \sum_{x=0}^{M-1} f(x)\, e^{-j2\pi ux/M}$$

$$f(x) = \sum_{u=0}^{M-1} F(u)\, e^{j2\pi xu/M}$$

$$F(u) = \frac{1}{M} \sum_{x=0}^{M-1} f(x)[\cos 2\pi ux/M - j \sin 2\pi ux/M]$$

$$|F(u)| = [R^2(u) + I^2(u)]^{1/2}$$

$$\phi(u) = \tan^{-1} \frac{I(u)}{R(u)}$$

2-D discrete function:-

$$F(u,v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y)\, e^{-j2\pi(ux/M + vy/N)}$$

$$f(x,y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u,v)\, e^{j2\pi(ux/M + vy/N)}$$

$$F(u,v) = [R^2(u,v) + I^2(u,v)]^{1/2}$$

$$\phi(u,v) = \tan^{-1}\left[\frac{I(u,v)}{R(u,v)}\right]$$

$$P(u,v) = |F(u,v)|^2 = R^2(u,v) + I^2(u,v)$$

advantages of frequence domain analysis.
> spatial is local frequency analysis is global
> Convolution with mask is just multiplication

(ii) filtering in frequency domain

pre processing $(-1)^{x+y}$ → fourier transform → filter function $H(u,v)$ → Inverse fourier transform → post processing

$H(u,v) \times F(u,v)$

$g(x,y)$ Enhanced image

Basic steps

1) multipli the input by $(-1)^{x+y}$ to center
2) Compute $F(u,v)$ → DFT
3) multiply $F(u,v)$ by filter
4) Inverse DFT
5) obtain the real part of image
6) multiply the result in (5) by $(-1)^{x+y}$

Idea ~~filter~~ Lowpass filter
> ringing effect in ideal image.

$$H(u,v) = \begin{cases} 1 & \text{if } (D(u,v) \le D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$$

gaussian → sharper at top then butterworth

$$H(u,v) = e^{-D^2(u,v)/2D_0^2}$$

$$P_T = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} P(u,v) \quad \alpha = 100 \left[\sum_u \sum_v P(u,v) / P_T \right]$$

$u \le$ radius (D_0), $v \le$ radius (D_0)

$$H_{hp}(u,v) = 1 - H_{LP}(u,v)$$

(#) Ideal highpass filter

$$H(u,v) = \begin{cases} 0 & \text{if } D(u,v) \le D_0 \\ 1 & \text{if } D(u,v) > D_0 \end{cases}$$

(#) Butterworth highpass filter

$$H(u,v) = \frac{1}{1 + [D_0 / D(u,v)]^{2n}}$$

(#) Gaussian highpass filter

$$H(u,v) = 1 - e^{-D^2(u,v)/2D_0^2}$$

Laplacian in frequency domain

$$T\left[\frac{\partial^2 f(x,y)}{\partial x^2} + \frac{\partial^2 f(x,y)}{\partial y^2}\right] = -(u^2+v^2)F(u,v)$$

$$\Downarrow$$

$$H_1(u,v) = -(u^2+v^2)$$

Subtract Laplacian from original image $\Downarrow$ frequency domain

$$g(x,y) = f(x,y) - \nabla^2 f(x,y)$$

$$g(x,y) = F(x,y) + (u^2+v^2)F(u,v)$$

$$H_2(x,y) = 1 + (u^2+v^2) = 1 - H_1(u,v)$$

Unsharp masking

$$f_{hp}(x,y) = f(x,y) - f_{lp}(x,y)$$

$$H_{hp}(u,v) = 1 - H_{lp}(u,v)$$

(#) High boost filtering

$$H_{hb}(u,v) = A - 1 + H_{hp}(u,v)$$

$$z(x,y) = \ln f(x,y) = \ln i(x,y) + \ln r(x,y)$$

$$Z(u,v) = F_i(u,v) + F_r(u,v) \qquad \text{DFT}$$

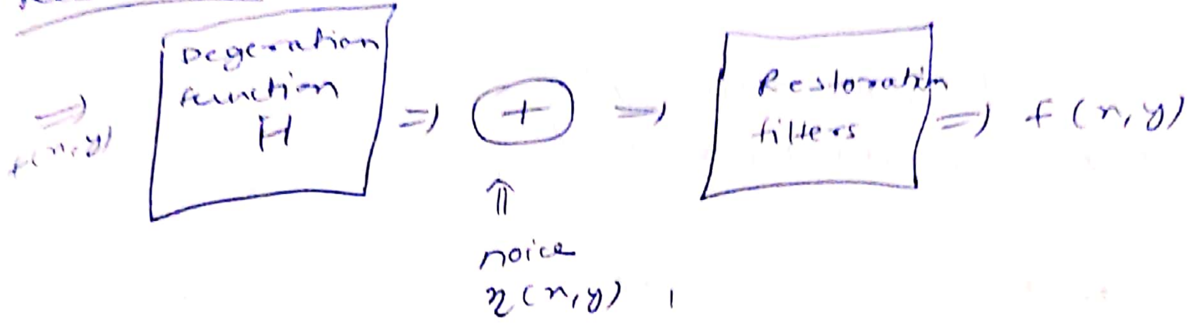
$$S(u,v) = H(u,v)\, Z(u,v) \qquad H(u,v)$$

$$s(x,y) = i'(x,y) + r'(x,y) \qquad (\text{DFT})^{-1}$$

$$g(x,y) = e^{s(x,y)} = i_0(x,y)\, r_0(x,y) \qquad \exp$$

$$f(x,y) \Rightarrow \ln \Rightarrow \text{DFT} \Rightarrow H(u,v) \Rightarrow \text{DFT}^{-1} \Rightarrow \exp \Rightarrow g(x,y)$$

Restoration



Degradation | Restoration

$$g(x,y) = (h*f)(x,y) + \eta(x,y)$$

linear position invariant operator

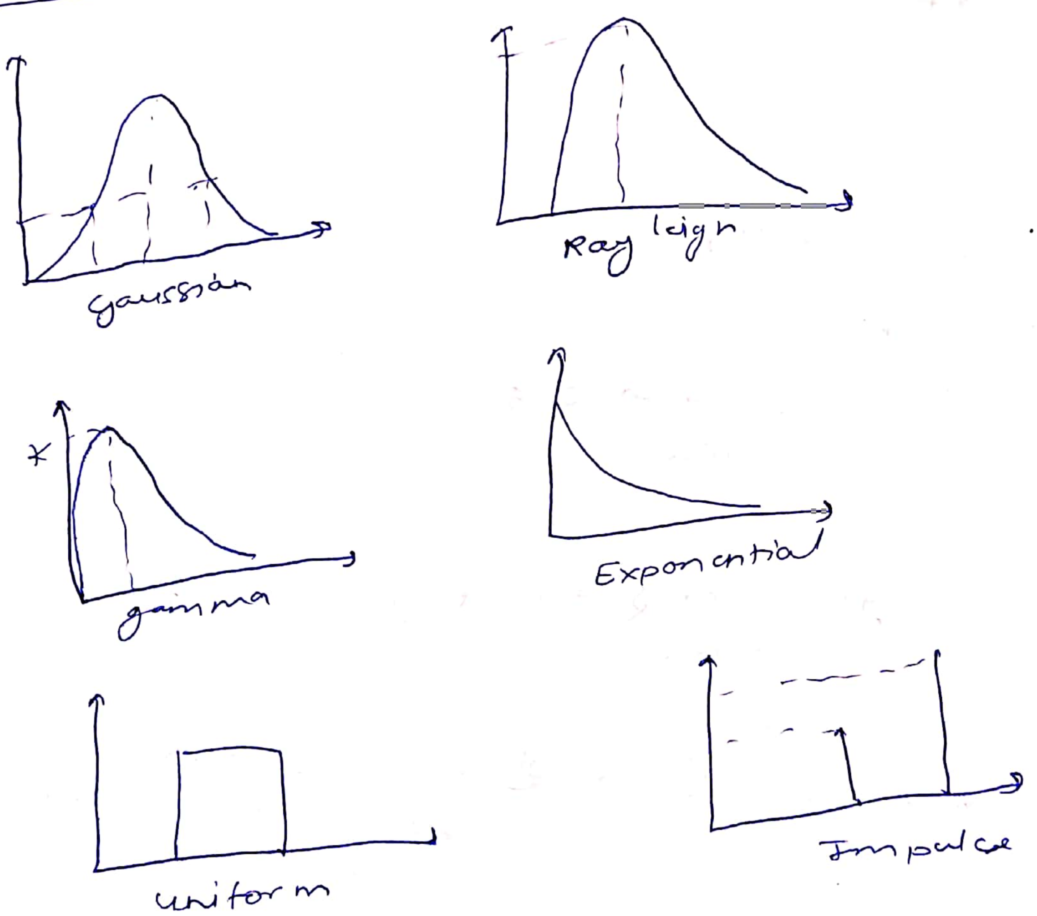
h(x,y) is degradation function

noice

$$G(u,v) = H(u,v)\,F(u,v) + N(u,v)$$

↳ frequency domain

noice models



(1) Restoring in the presence of noise only spatial filtering

$$g(x,y) = f(x,y) + n(x,y)$$

$$G(u,v) = F(u,v) + N(u,v)$$

(i) arithmetic mean filter

$$f'(x,y) = \frac{1}{mn} \times \sum_{(s,t)\in S_{xy}} g(s,t)$$

(ii) geometric mean filter

$$\hat{f}(x,y) = \left[\prod_{(s,t)\in S_{xy}} g(s,t)\right]^{1/mn}$$

(iii) Harmonic filter

$$f'(x,y) = \frac{mn}{\sum_{(s,t)\in S_{xy}} 1/g(s,t)}$$

(iv) contraharmonic mean

(v) order statics filter

↳ (i) median filter

$$f'(x,y) = \underset{(s,t)\in S_{xy}}{\text{median}} \{g(s,t)\}$$

(ii) max filter

$$f'(x,y) = \underset{(s,t)\in S_{xy}}{\max} \{g(s,t)\}$$

(iii) mid point filter

$$f'(x,y) = \frac{1}{2} * \left[\underset{(s,t)\in S}{\max} \{g(s,t)\} + \underset{(s,t)\in S}{\min} \{g(s,t)\}\right]$$

Scanned with OKEN Scanner

alpha trained mean filter

$$f'(x,y) = \frac{1}{(mn-d)} * \sum_{(s,t)\in S_{xy}} g_r(s,t)$$

Que:

1	7	5
6	2	3
1	4	2

① Arthmetic mean filter
② Geometric mean filter
③ Harmonic mean filter

Sol: ① $\frac{1}{9} \times (1+7+5+6+2+3+1+4+2)$

$\Rightarrow \frac{1}{9} \times 31 = 3.44 \Rightarrow 3$

② $(6\times 7\times 8\times 30)^{1/9}$

$\Rightarrow (5040)^{1/9} = 2.78 \Rightarrow 3$

③ $\frac{9}{(2+\frac{1}{6}+1+\frac{1}{7}+\frac{1}{5}+\frac{1}{3}+\frac{1}{4})}$

$\Rightarrow \frac{9}{3+0.16+0.14+0.2+0.33+0.25}$

$\Rightarrow \frac{9}{4.05} = 2.22$

Ex:

0	0	0	0	0
0	1	3	5	0
0	4	4	3	0
0	5	2	2	0
0	0	0	0	0

arthematic mean $\Rightarrow$

1	2	2
2	3	2
2	2	1

$\Downarrow$ geometric

0	0	0
0	3	0
0	0	0

0	0	0
0	3	0
0	0	0

HMF

b:

255	118	117
122	121	111
120	119	112

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

$\Rightarrow$ arithmetic mean filter

$121 \Rightarrow \frac{1190}{9} =$

geometric filter $\Rightarrow$

$\cancel{9.22 \times 10^{15}}$ $\cancel{9.22 \times 10^{18}}$ $(9.22 \times 10^{18})^{1/9} \Rightarrow 127$

Harmonic $\Rightarrow$ 124

Contraharmonic

$\theta = -2$ $\Rightarrow$ salt

$\theta = 2$ pepper

$\cancel{\frac{[1190]^{-1}}{[1190]^{-2}}} \Rightarrow 1190$

$\theta = -2$

$$\frac{\left(\frac{1}{255}\right) + \left(\frac{1}{118}\right) + \frac{1}{112} + \frac{1}{122} + \frac{1}{121} + \frac{1}{111} + \frac{1}{120} + \frac{1}{119} + \frac{1}{112}}{\left(\frac{1}{255}\right)^2 + \left(\frac{1}{118}\right)^2 + \left(\frac{1}{112}\right)^2 + \left(\frac{1}{122}\right)^2 + \left(\frac{1}{121}\right)^2 + \left(\frac{1}{111}\right)^2 + \left(\frac{1}{120}\right)^2 + \left(\frac{1}{119}\right)^2 + \left(\frac{1}{11}\right)^2}$$

$$= \frac{0.003 + 0.008 + 0.008 + 0.008 + 0.008 + 0.008 + 0.008 + 0.008}{0.000009}$$

Q) find the erosion image for the following

1	1	1	1	1	1
1	1	1	0	0	1
1	1	0	0	0	1
1	0	0	0	0	1
1	0	0	0	1	1
1	1	1	1	1	1

original image

6 x 6

1	0	1
0	0	0
1	0	1

structured element

(SE)

3 x 3

1s are background pixel do not match them

Central element ke basis pe match karna hai so need to match

* 0 *
0 0 0
* 0 *

so agar center element 1 hota toh

1 * 1
* 1 *
1 * 1

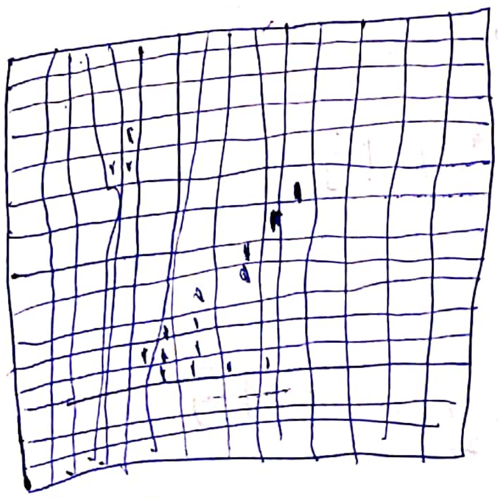
So compare karte 1 or solution me zero add hota

sol:

1	1	1	1	1	1
			0		
		0	0		

rest are all one

Ex
sol: 18



Ex

1	1	1	1
1	1	0	1
0	1	1	1
0	1	0	0

1	1
1	0

=>

0	1	0	0
0	0	0	0
0	1	0	0
0	0	0	0

offset nahi pata toh 0

ex

0	1	0
1	0	0
0	0	0

1	1
0	0

$\Rightarrow$

0	1	0
1	0	0
0	0	0

(1)

$\Rightarrow$

0	1	1
1	0	0
0	0	0

0	1	1	
1	0	0	0
0	0	0	0

$\Leftarrow$

0	1	1	1
1	1	0	0
0	0	0	0

$\Leftarrow$

0	1	1	1
1	1	1	0
0	0	0	0

$\Leftarrow$

0	1	1	1
1	1	1	1
0	0	0	0

(11) Erosion and dilation are dual of each other w.r.t. set complection and reflection.

$$(A \ominus B)^c = A^c \oplus \hat{B}$$

$$(A \oplus B)^c = A^c \ominus \hat{B}$$

$$(A \ominus B)^c = \{ z \mid (B)_z \subseteq A \}^c$$

$$= \{ z \mid (B)_z \cap A^c = \phi \}^c$$

$$= \{ z \mid (B)_z \cap A^c \neq \phi \}$$

$$= A^c \oplus \hat{B}$$

$$(A \oplus B)^c = \{ z \mid (\hat{B}_z)_z \cap A \neq \phi \}^c$$

$$= \{ z \mid (\hat{B}_z) \cap A^c = \phi \}$$

$$=$$

$$= A^c \ominus \hat{B}$$

perform opening and closing on below image using S.E.

1	1	1	0	1	1	1
1	1	1	1	1	1	1
1	1	1	0	1	1	1

Image (A)

1	1	1
1	1	1
1	1	1

S.E. (B)

⇒ opening → $(A \ominus B) \oplus B$

soln

$A \ominus B \Rightarrow$

0	0	0	0	0	0	0
0	1	0	0	0	1	0
0	0	0	0	0	0	0

$(A \ominus B) \oplus B$

1	1	1	0	1	1	1
1	1	1	0	1	1	1
1	1	1	0	1	1	1

closing

$(A \oplus B) \ominus B$

$A \oplus B \Rightarrow$

1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1

$(A \oplus B) \ominus B \Rightarrow$

1	1	1	1	1	1	1
1	1	1	1	1	1	1
1	1	1	1	1	1	1

Properties of opening and closing

(i) properties of opening

$(A \circ B) \circ B = A \circ B$

$A \circ B$ is a subset of A

If C is a subset of D then $C \circ B$

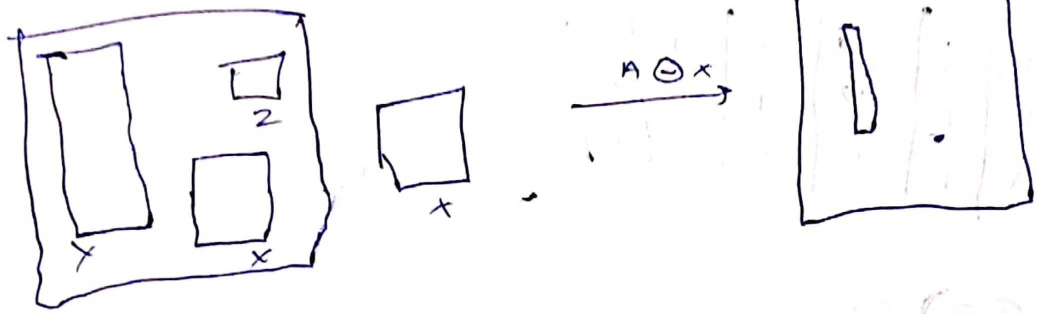
(ii) properties of closing

$(A \bullet B) \bullet B = A \bullet B$

A is subset of $A \bullet B$

(iii) Hit or miss

$$A \circledast B = (A \ominus B_1) \cap (A^c \ominus B_2)$$



Ex

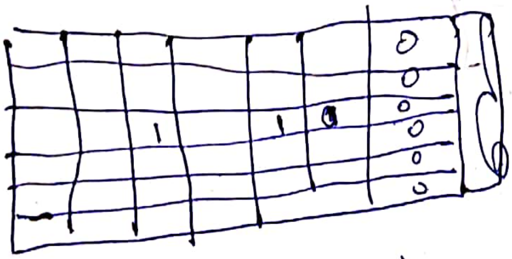
0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	1	1	1	1	1	0
0	0	1	0	1	0	0
0	0	0	0	0	0	0

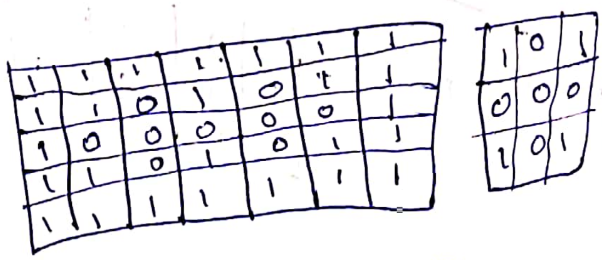
(A)

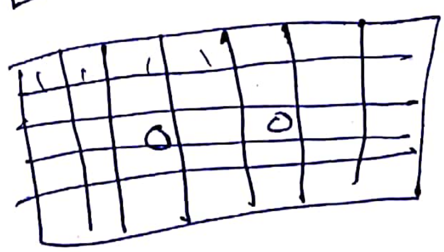
0	1	0
1	1	1
0	1	0

(B)

$A \ominus B$







Null matrix

Ex:

0	0	0	0	0	0	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	0	0	0	0	0	0

A

0	0	0
0	1	0
0	0	0

B_1

0	1	0
0	0	0
0	0	0

B_2

0	0	0	0	0	0	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	0	0	0	0	0	0

$A \ominus B_1$

1	1	1	1	1	1	1
1	0	0	0	0	0	1
1	0	0	0	0	0	1
1	0	0	0	0	0	1
1	0	0	0	0	0	1
1	0	0	0	0	0	1
1	1	1	1	1	1	1

A^c

0	0	0	0	0	0	0
1	1	1	1	1	1	1
0	0	0	0	0	0	1
1	0	0	0	0	0	1
1	0					1
1						1
1						1

1	1	1	1	1	1	1
1	1	1	1	1	1	1
1	1	0	0	0	1	1
1	1	0	0	0	1	1
1	1	0	0	0	1	1
1	1	1	1	1	1	1
1	1	1	1	1	1	1

and

$A^c \ominus B_2$

0	0	0	0	0	0	0
0	1	1	1	1	1	0
0	1	0	0	0	1	0
0	1	0	0	0	1	0
0	1	0	0	0	1	0
0	1	1	1	1	1	0
0	0	0	0	0	0	0

(11) Boundary Extraction

$$B(A) = A - (A \ominus B)$$

(12) Region filling / Hole filling

$$X_K = (X_{K-1} \oplus B) \cap A^c$$

Stop the iteration if $X_K = X_{K-1}$

(13) Extraction of connected component

$$X_K = (X_{K-1} \oplus B) \cap A$$

Let $D^i = X_K^i$

$$C(A) = \bigcup_{i=1}^{y} D^i$$

(14) Thining

$$A \otimes B = A - (A \circledast B)$$

$$A \otimes B = A \cap (A \circledast B)^c$$

(15) Thickening

$$A \odot B = A \cup (A \circledast B)$$

$$A \odot \{B\} = ((\ldots(A \odot B^1) \odot B^2) \ldots) \odot B^n)$$

(16) Skeleton

$$A = \bigcup_{K=0}^{K} (S_K(A) \oplus KB)$$

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Ex Hit miss transformation

0	0	0	0	0	0	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	0	0	0	0	0	0

A

0	0	0
0	1	0
0	1	0

B_1

0	1	0
0	0	0
0	0	0

B_2

$A \ominus B_1$

0	0	0	0	0	0	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

0	0	0	0	0	0	0
1	1	1	1	1	1	1
1	0	0	0	0	0	ϕ
1	0	0	0	0	0	1
1	0	0	0	0	0	1
1	0	0	0	0	0	1
1	0	0	0	0	0	1

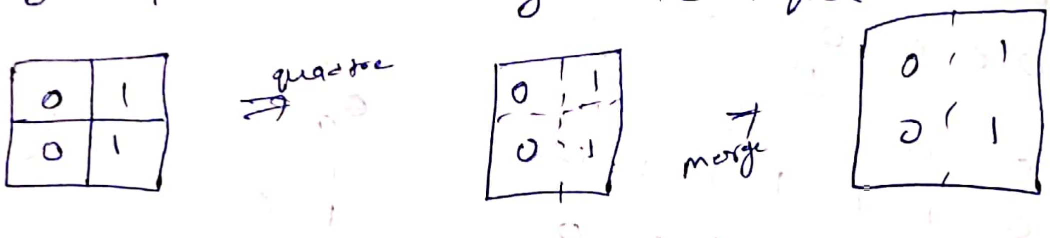
0	0	0	0	0	0	0
0	1	1	1	1	1	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

① Image segmentation

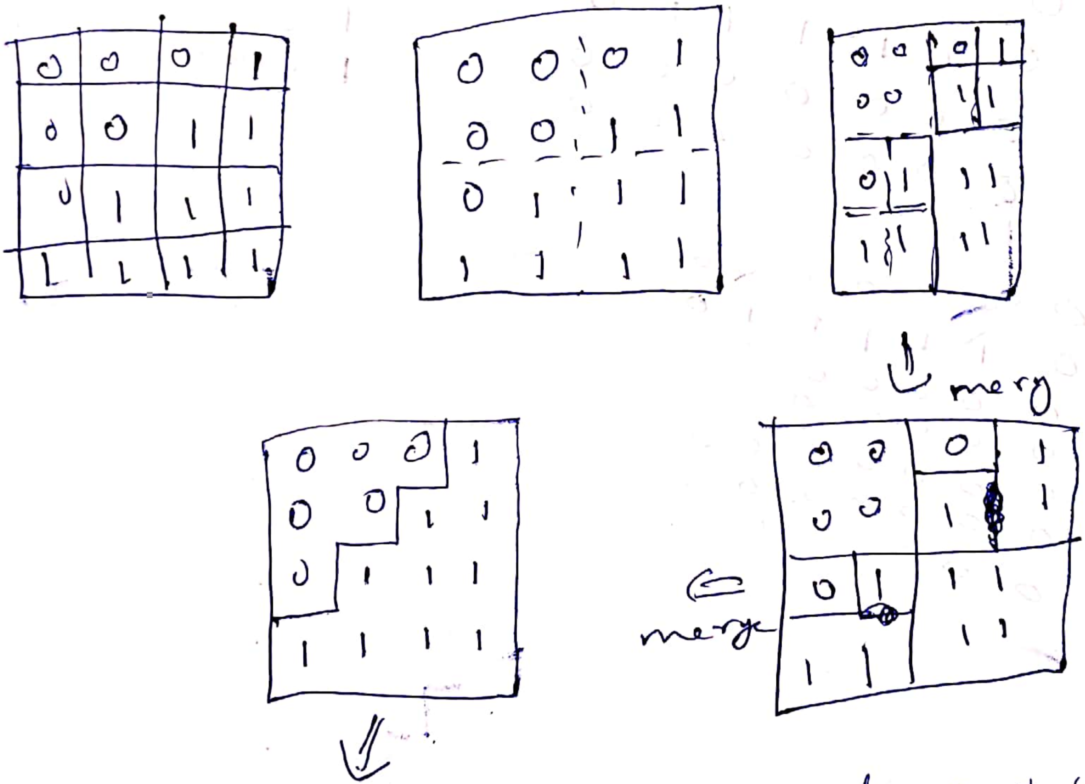
it refers to the process of partitioning an image into groups of pixels which are homogeneous with respect to some criterion

① Local segmentation
Global segmentation

② apply split and merge technique



ans apply split merg



zero 1
ones 0

In exam also need to draw quadtree

cluster Distance measure

(i) single cluster, complete cluster, average

minimum distance — maximum distance

$d(c_i, c_j) = \min\{d(x_{ip}, x_{jq})\}$

$d(c_i, c_j) = \max\{d(x_{ip}, x_{jq})\}$

Ques Do the clustering for the following using single link method

7 10 20 28 35

3 10 8 7

merge least distance

(7,10) 20 28 35

10 8 7

(7,10) 20 (28,35)

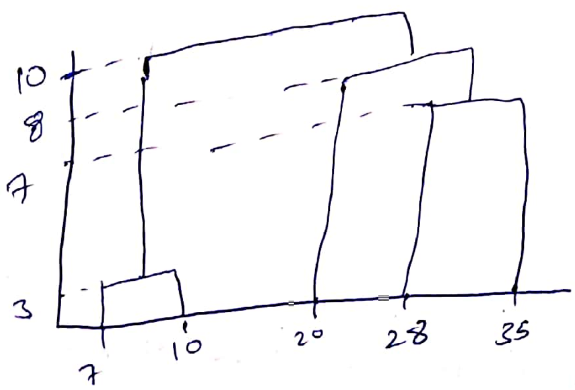
10 8

cluster I : (7,10)

cluster II : (20,28,35)

(7,10) (20,28,35)

10



(ii) complete link

7 10 20 28 35

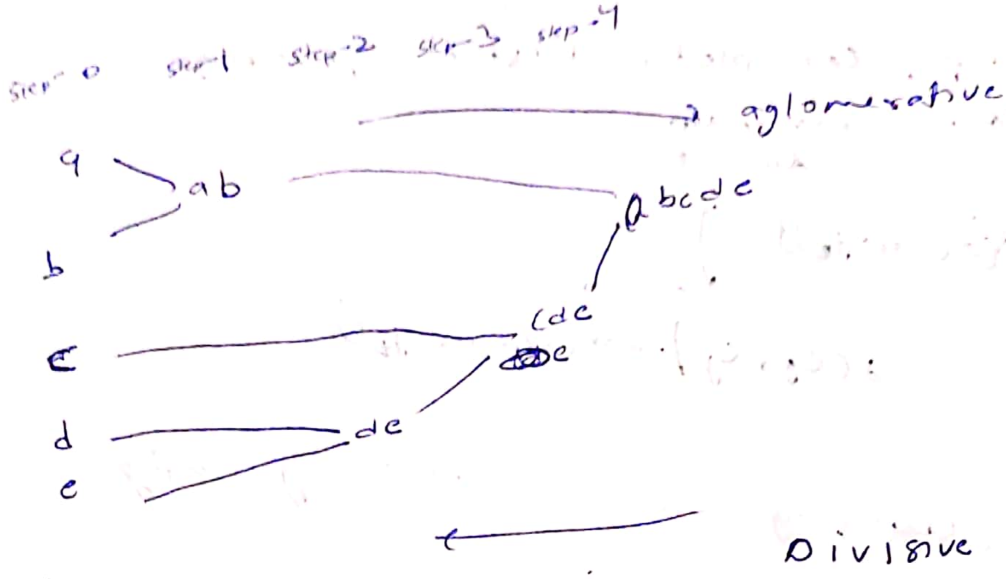
3 10 8 7

(7,10) 20 28 35

13 8 7

(7,10) 20 (28,35) ⇒ (7,10,20) (28,35)

27



classification of edges

(i) step edge

(ii) ramp edge

(iii) root edge

gradient operator

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

$$\phi(\nabla f) = \tan^{-1}\left[\frac{G_y}{G_x}\right]$$

(i) Edge strength, which is equal to the magnitude of the gradient

(ii) Edge direction, which is equal to the angle of the gradient

Roberts

-1	0
0	1

0	-1
1	0

Prewitts

-1	-1	-1
0	0	0
1	1	1

-1	0	1
-1	0	1
-1	0	1

sobel

-1	-2	-1
0	0	0
1	2	1

-1	0	1
-2	0	2
-1	0	1

$$\nabla \cdot \nabla = \begin{vmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{vmatrix} \cdot \begin{vmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{vmatrix}$$

$$\nabla \cdot \nabla = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \qquad \nabla \cdot \nabla = \nabla^2 = \text{Laplacian operator}$$

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) - 2f(x, y) + f(x-1, y)$$

also $$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) - 2f(x, y) + f(x, y-1)$$

drawback of laplacid operator

1) useful directional information is not available by the use of an laplacian operator

2) The Laplacian, being an approximation to the secound derivative doubly enhance any noice in the image

Laplacian of guassian

- The effect of noice can be minimised by smoothing the image prior to edge enhancement.

$h(r) = e^{-\frac{r^2}{2\sigma^2}}$ where $r^2 = x^2 + y^2$

σ : standard deviation

$$\nabla^2 h(r) = -\left[\frac{r^2 - \sigma^2}{\sigma^4}\right] e^{-r^2/2\sigma^2}$$

0	0	-1	0	0
0	-1	-2	-1	0
-1	-2	16	-2	-1
0	-1	-2	-1	0
0	0	-1	0	0

Hough transform

$(x, y) = 1, 2$ $\quad$ $x, y = 3, 4$

$y = ax + b \Rightarrow 2 = a + b$ $\quad$ $4 = 3a + b$

$a = 1, b = 1$

for $(x, y) = 1, 2$

~~y — m~~

$y = mx + c$

$c = -m(1) + 2$

$c = -m + 2$

$c = 0 \quad m = 2$

$m = 0 \quad c = 2$

for $(x, y) = 3, 4$

$4 = 3m + c$

$c = 3m - 4$

$c = 0 \quad m = 4/3$

$m = 0 \quad c = +4$